

**REAL-TIME IMAGING FOR COVID-19 &  
BREAST CANCER DETECTION**

**A PROJECT REPORT  
submitted in partial fulfillment of the  
requirements for the award of**

**BACHELOR OF TECHNOLOGY**

**in**

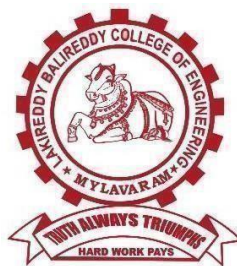
**ELECTRONICS & COMMUNICATION ENGINEERING**

**By**

**D. KALYAN KUMAR**

**(22761A04E4)**

*Under the Guidance of*  
**CH. SIVA RAMA KRISHNA**  
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**DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING  
LAKIREDDY BALIREDDY COLLEGE OF ENGINEERING  
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**L.B.Reddy Nagar, Mylavaram – 521 230.**

**Affiliated to JNTUK, Kakinada & Approved by AICTE, New Delhi**

**Accredited by NBA (under tier-I) and NAAC**

**Certified by ISO 21001:2018**

**2025- 2026**

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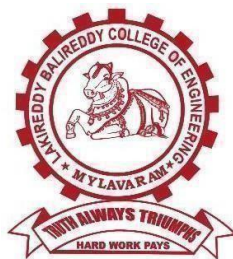
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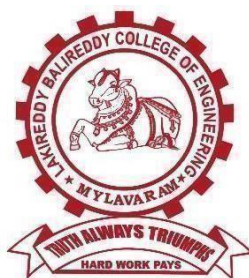
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## **CERTIFICATE**

This is to certify that project work entitled “**REAL-TIME IMAGING FOR COVID-19 & BREAST CANCER DETECTION**” is a bonafide work done and submitted by **D.KALYAN KUMAR (22761A04E4)** in partial fulfillment of requirement for the award of Bachelor of Technology in Electronics and Communication Engineering in Lakireddy Bali Reddy College of Engineering, Mylavaram during the academic year 2025-2026.

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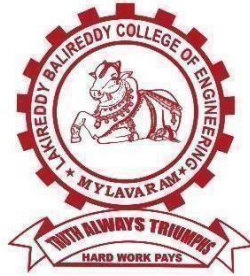
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**DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING**



**DECLARATION**

We hereby declare that the project entitled **“REAL-TIME IMAGING FOR COVID-19 & BREAST CANCER DETECTION”** submitted in partial fulfillment of the requirement for award of Bachelor of Technology in Electronics and Communication Engineering, Mylavaram, is our original work and has not submitted to any other institution or University for the award of any degree. This work was carried out during the academic year **2025-2026**.

**D.KALYAN KUMAR**

**(22761A04E4)**

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## **Abstract**

Medical imaging plays a crucial role in disease diagnosis by enabling non-invasive analysis of internal structures. This paper presents a deep learning-based framework for automated classification of COVID-19 using chest CT images and breast lesions using ultrasound imaging. EfficientNetB0 with transfer learning is employed for COVID-19 detection, while a lightweight Convolutional Neural Network (CNN) is used for breast cancer classification into benign, malignant, and normal categories.

The system follows an end-to-end pipeline including preprocessing, training, inference, and deployment through a Flask-based web application that provides real-time predictions with confidence scores. Edge detection techniques are incorporated for basic interpretability, and a SQLite database is used for logging results.

To further improve model robustness and reduce overfitting, data augmentation techniques including rotation, flipping, and scaling are used. To guarantee dependability and consistency across datasets, the models are assessed using important performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. To boost performance, hyperparameter tuning is used to tune parameters like learning rate, batch size, and epochs.

Healthcare practitioners can input medical photos and get immediate diagnostic predictions because to the system's user-friendly interface. Faster processing and adaptability for deployment in contexts with limited resources are guaranteed by its lightweight architecture.

## **Keywords**

- Deep Learning in Medical Imaging
- COVID-19 and Breast Cancer Classification
- Real-Time Diagnostic System

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# **CHAPTER 1**

## **INTRODUCTION**

Medical imaging has been an important component of contemporary healthcare to diagnose diseases early and accurately. With the rapid development of artificial intelligence and deep learning technologies, automated analysis of medical images is becoming increasingly popular. In particular, real-time imaging systems allow making the diagnosis faster and making immediate clinical decisions, which is essential in severe diseases such as COVID-19 and breast cancer.

The COVID-19 is a highly contagious respiratory illness that predominantly affects the lungs and can be successfully identified with the help of CT scans in the chest. Early diagnosis will enable prompt treatment and reduce chances of serious complications. Likewise, breast cancer is a major cause of death among women around the world, where early detection via ultrasound imaging can greatly enhance survival rates.

Conventional diagnostic processes use a lot of professional radiologists. Such a method may be time consuming and subject to human error particularly in a high clinical workflow. To address these challenges, deep learning techniques provide automated, precise and efficient processing of medical images. The real-time systems can detect and classify the abnormalities with high accuracy by utilizing convolutional neural networks (CNNs) and transfer learning models.

The current project will focus on creating a real-time imaging system, which will integrate deep learning models to identify COVID-19 in CT images and breast cancer in ultrasound scans. The system aims to offer fast, reliable, and easy-to-access diagnostic assistance, particularly in resource-constrained environments, thereby enhancing healthcare outcomes and reducing the diagnostic burden.

The presented system aims at the ease of use and scalability of the proposed system with a web-based interface to be accessed by different healthcare centers. It also provides safe storage and handling of patient information to be analyzed and tracked in future. The system assists clinicians to make faster and more informed decisions by adding the elements of real-time predictions. All in all, this method increases the effectiveness of diagnoses and enhances patient care and treatment outcomes.

## 1.1 Introduction to Visual Studio Code

VS Code or Visual Studio Code is a lightweight and open-source code editor developed by Microsoft. It is also used by many developers to write, debug, and manage code in any of the programming languages. VS Code has gained popularity since its introduction due to its simplicity, flexibility and powerful nature. It allows many types of languages such as Python, Java, C++ and JavaScript among others.

Its ability to be extended is one of its strong points. VS Code can be enhanced by installing add-ons via the VS Code Marketplace. These extensions include support tools such as language support, debugging, code formatting, version control integration and help provided by AI. The editor has a user-friendly user interface with such features as syntax highlighting, code completion (known as IntelliSense), error checking, and debugging facilities. The IntelliSense helps the developers with the code completions, function definitions, and information about the variables which makes the code more efficient and makes fewer mistakes.

The other important aspect of VS Code is that it is integrated with Git. This allows developers to control versioning right in the editor. It also makes it easy to monitor the change, work with colleagues, and maintain the project history without the necessity to use other tools. Also, VS Code has a powerful terminal that enables developers to run commands, run scripts and manage project environments within the same interface. This combined terminal can enhance the efficiency of workflow and eliminate the usage of different command line tools.

The editor is very user-friendly and customizable, allowing users to customize themes, keyboard shortcuts and settings to their preferences. It has light and dark themes to guarantee comfort when used over a long period. Besides, VS Code is cross-platform, which means that it can be used with Windows, macOS, and Linux operating systems. Its lightweight nature guarantees fast execution, even on low-resource computers, and is compatible with amateurs and expert programs.

To conclude, Visual Studio Code is a robust and user-friendly development environment that can be used with ease and have great capabilities. Its popularity in the software

development industry has made it an essential component in the coding, debugging, and project management sectors throughout different areas.

## **12 Importance of Medical Imaging in Modern Healthcare Motivation**

Modern healthcare is indeed propelled by medical imaging. Physicians use it to diagnose, follow-up, and treat any type of conditions. They can see inside the body without any surgery; with things such as CT scans and ultrasounds. This means that they will be able to identify and resolve issues earlier, make smarter decisions, and intervene sooner, should a patient require assistance.

However, despite the technology being amazing, it has given rise to some headache. The amount of imaging data keeps exploding. Scans are simply piling up and there are not enough radiologists to match. Individuals desire fast, precise responses and reading these pictures is time-consuming. Errors can creep in, and even the old professionals do not always see eye-to-eye.

This is why, there is great interest on computer-aided diagnostic systems. These instruments serve as new glasses-always sharp, always alert and never fatigued. Convolutional Neural Networks and deep learning in general is a huge advancement in this regard. Such networks identify complex, subtle patterns in medical images-thing that a human being may not notice. They are also fast and they do not require a lot of hand-holding.

Directly applying deep learning to real time imaging changes the speed and quality of clinic operation. It allows doctors to have an easier load, maintains more consistent results, and delivers quicker diagnoses. Besides, operating them using easy-to-use web or cloud-based solutions implies that even small clinics or rural hospitals receive the best tools. Ultimately, it all leads to improved patient care, more effective hospitals and an actual step towards precision.

### **13 Challenges in COVID-19 and Breast Cancer Imaging Existing Model**

The world became enlightened about the need to have automated diagnostic systems when COVID-19 hit. Hospitals were full, physicians were overburdened, and the virus was spreading faster than the test results could keep up. RT-PCR tests-they are the gold standard in diagnosing the virus-but there was always a delay and shortage. Plus, the results weren't always conclusive.

Chest CT scans came to the rescue, there. They turned into an important reserve. CT tests provided a better picture of what was wrong with the lungs, were able to see the progression of pneumonia, and were able to monitor the level of badness. However, radiologists were overwhelmed with so many scans. The necessity of automated tools that would assist in sorting and flagging urgent cases became unavoidable.

Although COVID-19 has brought these issues to the frontline, breast cancer has been a significant problem on the international front. It is always the timing, but the equipment is important. Breast ultrasound is trendy-it is cheap, it is available in most clinics and it is safe as it does not have any radiation. Ultrasound is particularly useful in dense breast tissue, where mammograms find difficulties. The downside? Ultrasound images can be tricky. They are loud, usually shaggy and the outcome varies greatly depending on who is holding the probe. Sometimes the tumor edges are so delicate, even professionals are not certain that they see anything.

This is where deep learning comes in. Using strong CNN-based algorithms on a ultrasound image, systems can identify tiny details-textures, shapes, edges-which a human may miss. All of a sudden, diagnosis becomes more precise and much more consistent. Introducing artificial intelligence in medical imaging accelerates all that. It reduces the discrepancies that exist among physicians regarding their judgments and decisions become much more accurate. These devices are not here to substitute doctors and only to support them and simplify their work.

Better still, you can unite these methods-COVID-19 CT scans and breast cancer ultrasound-in a single deep learning model. It truly demonstrates how scalable and flexible these AI systems can be to address the most significant issues in medicine.

## 1.4 Role of Deep Learning and Proposed Framework

COVID-19 CT scans as well as breast ultrasound pictures are both associated with their own set of challenges, which makes them actually good subjects to machine learning and deep learning methods. Deep learning-which is essentially a subset of artificial intelligence-has performed quite well in image recognition since it learns what is important directly off the pixels,<sup>4</sup> rather than relying on humans to pre-write features. Deep learning models can identify intricate and subtle patterns in images without human-imposed cues, instead of the latter. That normally implies improved accuracy and more dependable outcomes.

A potent method of deep learning is known as transfer learning. In this case, you will use models that already have been exposed to a large amount of data and then make them specific to a particular task-like medical imaging. This is applicable to medicine particularly where one cannot get a plethora of labeled images. It does not require as many new samples to get good results since it starts with pretrained models. It accelerates the process, makes the model learn more meaningful features and makes it even more accurate even in the case of small dataset.

In terms of identifying COVID-19, such models as EfficientNet have already demonstrated a high level of performance in identifying the symptoms of a disease in CT images. Conversely, in case of breast ultrasound, you are able to develop lightweight convolutional neural networks (CNNs) capable of handling grayscale scans and categorizing them as such: benign, malignant, or normal. These models emphasize learning the details that matter-shape, texture, the edges of lesions-and that can eliminate much of the subjectivity you can receive by reading manually.

Nevertheless, despite all this development, a huge gap remains. Clinics do not necessarily translate research models into practice. There are several systems which are hard to integrate, not easy to interpret and not ready to scale up. The paper addresses these issues by presenting an entire end-to-end imaging system that applies deep learning to COVID-19 detection and breast lesion diagnosis. The system assembles standardized preprocessing, solid training techniques, and deployment-ready design with a simple Flask web interface.

It also offers the score of confidence and elaborate visualizations-right on the edge-to make the results easier to interpret. In addition, it records the results of inference in a SQLite database, which allows one to easily monitor the results and continue to improve. The imaging AI tools can be transferred out of the laboratory and into real-world clinics by staying in a single pipeline-preprocessing, model predictions, visualization, and logging-the-framework, particularly in cases where resources are limited.

## **CHAPTER 2**

### **LITERATURE SURVEY**

#### **2.1 Edge Detection Techniques in Medical Imaging**

The article by Jarjees and his colleagues in 2025[1] introduces a system that is based on a type of computer chip known as an FPGA to identify edges on medical images, in this case, on the images of chest X-rays, to aid in diagnosing COVID-19.

The authors believe that it is quite necessary to promptly and identify issues in the lungs, thus allowing the doctors to diagnose the COVID-19 at an early stage.

They developed this system since the conventional approaches such as Sobel and Prewitt do not perform well in locating the edges when the image is not in focus.

Their system employs logic in searching the edges of an image and is effective in handling images that are not clear.

They made the system work very fast by using a computer program known as MATLAB and a special hardware platform the FPGA board.

The system employs math functions known as Gaussian and trapezoidal membership functions to ensure that it locates edges precisely.

When they experimented with the system they discovered that it was effective with high levels of similarity between the output of the computer program and the hardware platform and was also effective at identifying edges in images of infected lungs.

The system is also good in that it does not require a lot of computer power to operate as opposed to some of the methods.

But this system can only be effective with X-ray images. Has difficulties with other types of images such as CT and MRI scans.

On the whole this study contributes to the diagnosis within a short time and with high precision through the application of fuzzy logic and special computer chips.

The Edge Detection Techniques in Medical Imaging such as the one that is presented in this study are actually crucial in the diagnosis of COVID-19 through the use of chest x-ray images.

The Edge Detection Techniques, in Medical Imaging, aided with logic and hardware acceleration, assist the COVID-19 diagnosis.

## **2.2 Electromagnetic Methods in Breast Cancer Diagnosis**

In 2011[2], Hassan and El-Shenawee have conducted a review on the use of electromagnetic methods in detecting breast cancer. They examined how they could identify breast cancer other than the routine X-ray mammography. They discussed the issues with the techniques. Such issues are getting results and health risks.

The article examines such electromagnetic techniques as microwave imaging and electrical impedance tomography. It also looks at diffuse tomography and microwave radiometry. Additionally it looks at -potential detection and magnetic resonance imaging. All these approaches are considered in detail.

Microwave imaging and radiometry are good, the authors believe, since they are able to distinguish between malignant tissues. This implies that they are able to detect tumors without subjecting an individual to harm. Electrical impedance tomography and diffuse optical tomography are also good since they can indicate how the tissues are functioning. They accomplish this by gauging the passage of electricity and light through the tissues.

Magnetic resonance is a high-tech technique. It is quite effective in demonstrating what is happening in the body. It is however also very costly. The authors believe that one of the approaches to detecting breast cancer at an early stage is through electromagnetic techniques. They do not use radiation, which's a plus. However there are still some problems. These problems include not being able to see the tissues clearly and being very complicated.

The authors are hoping that someday these techniques will be applied in hospitals to assist people. They believe that through research we are able to perfect these methods and make them more useful. This will aid doctors to identify breast cancer earlier and easily. Electromagnetic techniques, such as microwave imaging and electrical impedance tomography will come in handy. The future of breast cancer diagnosis lies in electromagnetic.

## **2.3 Risk Factors and Prevention of Breast Cancer Overview**

Sun and their team (2017) [3] looked into the things that can increase the risk of breast cancer and ways to prevent it. They discovered that a combination of genes, hormones, environment and life style preferences affect breast cancer. Aging, having a family history of breast cancer, being overweight, drinking alcohol and having children in life can be the risk factors.

The researchers stressed that finding breast cancer early and making lifestyle changes can help lower the risk. They proposed exercising and eating regularly and avoiding things that may put them at a higher risk of the disease. They also discussed the influence of hormones and some genetic alterations such as BRCA1 and BRCA2 on breast cancer.

The reason why this study is important is that it enables us to comprehend the risk of breast cancer which is the key in developing good diagnostic systems. It demonstrates the importance of screening early and promotes creating detection strategies with artificial intelligence and imaging technologies.

The results of the research justify the necessity, of check-ups and healthy living to minimize the risk of breast cancer.

Knowing these risk factors will allow us to make preventive measures against breast cancer.

## **2.4 Cadherins and Catenins in Breast Cancer**

The research by Cowin and others from 2005[4] looks at how cadherins and catenins affect breast cancer. These are molecules which assist cells to stick together. They play an important role in maintaining healthy cells and tissues. When they fail to perform properly it may cause cancer and tumors to spread.

In this study it was discovered that in many cases when E-cadherin is lost or mutated it is associated with spread breast cancer. The cells are free to move freely when they are not sticking together as they are supposed to in normal cells. The paper also discusses the interaction of cadherins with the signals within cells.

This book lets us know how cancer develops at that level. It provides us with concepts of how molecule changes can influence what we observe in pictures. It can be used to make

computer programs to locate tissue structures within images of breast cancer. Cadherins and catenins play a role, in this. Breast cancer is a disease and knowledge of cadherins and catenins can help us fight it.

## **2.5 Breast Cancer Histopathology Image Analysis: A Review Overview**

The researchers Veta and his team looked at how computers can help analyze pictures of breast cancer tissue in a study from 2014[5]. They wanted to see if computers could help doctors diagnose breast cancer by looking at pictures of tissue under a microscope. Breast cancer diagnosis is the focus of this study. The team used computer programs to analyze these pictures.

They talked about the ways to make these computer programs work like making the pictures clearer and finding the important parts. They also said that it is not easy to make these programs work because the pictures can look different and the tissue can be complicated.

This study is very important for people who are working on making computers smarter. It shows that computers can help doctors look at pictures and make a diagnosis. It also says that using kinds of computer programs like CNNs can help make diagnoses more accurate. Breast Cancer Histopathology Image Analysis is what this study is, about. Breast Cancer Histopathology Image Analysis is a deal because it can help doctors diagnose breast cancer better.

## **2.6 Review of Electromagnetic Techniques for Breast Cancer Detection**

In one study, Hassan and El-Shenawee (2011) [6] examined the imaging techniques as an alternative method of detecting breast cancer. They were interested in determining how they could do this as compared to conventional mammography.

The researchers investigated such methods as microwave imaging and electrical impedance tomography and MRI. These methods are sensitive to detecting tissue properties. Nonetheless they too have certain issues such as not being extremely clear and consuming much computer power.

This article has great significance as it describes the theory of these imaging technologies.

It demonstrates that to enhance traditional imaging systems, we should apply computer techniques such as deep learning. The electromagnetic techniques are used to detect breast cancer. Electromagnetic techniques, for breast cancer detection are still being researched.

## **2.7 Microwaves for Breast Cancer Detection Overview**

This research[7] was done by the individuals who investigated using microwaves to detect breast cancer in 2003, Fear and his team. Their interest was to determine whether they could use microwaves to distinguish between tissue and cancer tissue.

They did this through microwaves which they used to produce images of the breast. This can be done by doing so since it is painless and it does not involve any radiations. They found out that microwaves can be used to find tumors. They experienced certain issues with obtaining clear pictures and interference.

The current research demonstrates that new image making can be of great assistance. It is significant as well since it may assist us in learning how to apply types of data with artificial intelligence to arrive at diagnoses that are more accurate. Microwaves, in the detection of breast cancer is a still developing technique. The research on microwave detection of breast cancer is interesting.

## **2.8 Breast Cancer Detection Using Extreme Learning Machine with**

Wang and his team wrote a paper in 2019[8] that talks about a way to detect breast cancer.

They used a combination of a kind of computer vision called CNN-based feature extraction and something called Extreme Learning Machine for classification.

The Extreme Learning Machine is really good at classifying things because it uses features that are extracted from the CNNs.

When you combine all these features together it makes the classification a lot better.

I think this paper is really important for your project because it shows that using learning with traditional classifiers is a good idea.

It also says that using EfficientNet and CNN architectures is a way to go.

The paper is about Breast Cancer Detection Using Extreme Learning Machine. It is relevant, to your project.

## **2.9 Thermal Imaging and Deep Learning for Breast Cancer Detection**

Roslidar and others (2020) [9] look at using imaging with deep learning to find breast cancer.

Thermal imaging finds temperature changes from tissue activity.

Deep learning models make detection more accurate by spotting patterns.

This study shows how non-invasive imaging and AI can help find breast cancer

It backs up the idea of using time diagnostic systems.

Thermal imaging and deep learning are tools for breast cancer detection.

They help doctors find breast cancer early.

Deep learning models are good, at finding patterns in images.

Thermal imaging is a -invasive way to find temperature changes.

## **2.10 Particle Swarm Optimization for Breast Cancer Prediction Overview**

Sakri and other people did a study in 2018[10] where they used Particle Swarm Optimization for picking the features in breast cancer prediction models.

They used Particle Swarm Optimization for this.

Particle Swarm Optimization helps to pick the features for breast cancer prediction models.

This helps the models to be more accurate. It also helps to make the process simpler.

The Particle Swarm Optimization method makes the models work better.

This study shows that using techniques to make things better is important for machine learning models.

It is also useful for making deep learning work better.

The Particle Swarm Optimization is helpful, for breast cancer prediction models.

## **2.11 Deep Learning for Histological Diagnosis of Breast Cancer Overview**

The researchers Yari et al. Did a study in 2020 [11] that looked at how deep learning can be used to analyze pictures of breast cancer tissue.

Deep learning is really good at looking at pictures and figuring out what is going on.

CNN models are very good at telling if a tissue is cancerous or not.

This study says that having a lot of pictures to look at and making sure the computer is trained properly is very important.

This research says that deep learning is a tool for looking at medical pictures and it agrees

with using a CNN model for this kind of work.

Deep learning, for breast cancer diagnosis is what this study is about and it shows that deep learning can be very helpful.

## **2.12 Deep Learning Method for Breast Cancer Classification Overview**

Liu et al. In 2022[12] suggested a learning method to classify breast cancer from pathology images.

They used this method to get results.

The model works with different types of images because it uses strong neural networks and good image preparation techniques.

This study shows that deep learning is useful in diagnosis and helps in creating systems that can automatically classify breast cancer.

Deep learning methods, like this one can make breast cancer classification more accurate.

## CHAPTER 3

### SOFTWARESUSED

In order to create & deploy a medical imaging (MI) based decision support system, an suitable software environment should be created that fosters effective functioning, scalable and reproducible outcomes and usability. The aim is to create a universal integrated system of COVID-19 detection based on Deep Learning models and breast cancer classification. This multi-faceted nature of programming languages, libraries, frameworks and databases will be required to achieve the goals outlined. This section will give a detailed description of the software requirements that should be taken into consideration in order to have successful design and operation.

#### 3.1 Programming Language and Development Environment

The main section of the program is coded in Python, now the most popular language of artificial intelligence and computing. Python is quite a popular language among scientists working in the field of computing and it has numerous data manipulation tools, neural networks, and numerous others as you can discover through its numerous scientific libraries. It possesses several main advantages or characteristics like easy to program, simple syntax (e.g. keywords are written in English), easy to organize (e.g. modules/libraries are organized in the same way as other programming languages) and high number of users. These characteristics render Python a great development/resource in developing a prototype and in production. Along with Jupyter Notebooks, Google Colab, or IDEs (Integrated Development Environments) such as PyCharm and Visual Studio Code to assist with programming (debugging, visualizing, building), it is also strongly suggested to use virtual path(s) (e.g., venv, conda) to manage dependencies across systems to ensure that your models can be reproducible across machines.

#### 3.2 Deep Learning Frameworks

Deep Learning frameworks play an essential role in the model development, training and inference. In particular, the following frameworks make up the primary components of AI algorithms: TensorFlow - TensorFlow is the open-source deep learning framework created by Google, which has become widely accepted for producing and delivering machine

learning models. TensorFlow has flexibility in terms of model design and offers capabilities for both research and production applications. Keras - Keras is the high level API developed by TensorFlow to facilitate the rapid development of neural networks. Neural networks have been simplified for creating complex architectures using CNNs and other methods such as EfficientNet (transfer learning). Also, these frameworks allow for efficient development of COVID-19 CT classification models from transfer learning and breast ultrasound classification models using custom developed CNN architectures. Additionally, both frameworks provide GPU accelerated processing, which greatly decreases the training time to learn from large datasets.

### **3.3 Image Processing Libraries**

To begin, preprocessing medical images is an important part of assuring model performance and consistency. This section discusses the libraries commonly used for processing images and preparing them for a model:

- OpenCV: Used for resizing, normalizing, filtering, and detecting edges using the Canny or Sobel operators.
- Numpy: Provides support for numerical operations and array operations which are important for working with image data.
- Pillow (PIL): Used for loading images and changing their formats.

These libraries allow for specific types of preprocessing to be done on CT and ultrasound imaging to ensure that deep learning models will have a standard input format. The edge detection techniques from OpenCV will also help with explainability of models as it can provide evidence of structural features within an image.

### **3.4 Web Framework for Deployment**

In constructing a user-friendly and easily accessible system, a lightweight web application is developed from the Flask framework. Flask is a micro web framework written in Python, enabling the rapid development of web-based interfaces with little overhead. Flask can be used to integrate machine learning models into a real-time application.

The Flask-based interface includes the following functionalities:

- The ability to upload images of CT and ultrasound scans.
- Select the model to be used (COVID-19 or classification of breast cancer).
- The ability to perform predictions and inference in real-time.
- Displaying the confidence score and probability distribution of a particular prediction.
- The ability to visualize the image after edge detection using Canny and Sobel edge detectors.

Due to the low complexity and modularity of Flask, it is a good choice for deploying this system in a resource-constrained environment while also providing room for scalability when adding additional features.

### **3.5 Database Management System**

SQLite is an easy-to-use relational database management system, which means it doesn't need a server or any kind of heavy database software to run. SQLite can be used when you need to save data locally (for example: in your own computer or cell phone), but do not wish to have to manage a large database server to do so. The SQLite database is designed to hold the following information:

- The file paths of uploaded images
- The predictions of the models, in the form of class labels.
- The confidence and probability value of the models' predictions.
- The time the inference was made.
- Image(s) that have been processed (in the form of edge maps).

This means of logging data permanently allows you to see how well the system is working, as well as provide a means to analyze errors and also allow monitoring of the model's performance over time. In addition, the SQLite database allows for the creation of dashboards and analytic tools to help assess and understand the performance of the models over time.

### 3.6 Supporting Libraries and Tools

There are a number of other libraries that are not critical to building a framework but can enhance a developer's ability to develop a framework or system. An example of such libraries are:

- Matplotlib and Seaborn for visualizing the training and validation accuracies, loss curves and performance metrics.
- Scikit-learn to provide evaluation metrics such as accuracy, precision, recall, confusion matrix and classification reports.
- Pandas to assist with the management of structured data and logs during analysis.

These libraries and tools will allow you to:

- Evaluate the model.
- Visualize the model.
- Document the data.

Making it possible to follow a complete and consistent development process

### 3.7 System Compatibility and Execution Requirements

This software solution will be developed so that it works across platforms regardless of an operating system (e.g. Windows, Linux, macOS). To achieve the best results from the software when using it for training machine learning models it is highly recommended to use:

- At least Python version 3.8
- At least TensorFlow version 2.x
- Minimum of 8 GB of RAM (16 GB recommended)

Use of an NVIDIA GPU with CUDA support for improved performance (optional). Once the system has been trained use for the purpose of using it within clinics and rural healthcare facilities will not require any special hardware (i.e. it can run on standard CPUs).

### 3.8 Security and Reliability Considerations

Data security and system reliability are essential in all medical systems. The software design of the present implementation emphasizes functionality and provides an opportunity

for implementing additional security features such as:

- User authentication and role-based access control
- Secure handling of files and encrypted storage
- Input validation to prevent the upload of malicious data

In addition, the system's modular architecture provides for updating or replacing independent elements (models, databases, interface) without impacting the overall functionality of the system.

### **3.9 Scalability and Future Extensions**

The architecture of the software can be built to grow and change with time and future technology progress. Future capabilities that could come into play include:

1. Plugin capabilities where other sophisticated machine learning techniques (e.g., Grad-CAM) can be added to increase the ability to explain how the model arrived at the decision.
2. Additional imaging modalities, such as X-rays and CT/MRI scans, can be integrated.
3. Deployment in cloud computing systems for remote access and the ability to adjust based on internet connectivity.
4. The application of federated learning for privacy-preserving training of the model. Based on the architecture's usage of existing frameworks and standardized approaches, it provides for changes with the implementation of new technologies or continuities in the clinical setting.

## CHAPTER 4

### METHODOLOGY

This system unites a deep learning system to classify medical images, and two key healthcare tasks: identifying COVID-19 in the chest CT scans and classifying the ultrasound images of breast lesions. The whole assembly is a seamless, end-to-end. pipeline- right after data prep and preprocessing, to designing and training the model, justifying its decisions, and in fact implementing it in actual practice. Here's a closer see what occurs in every step of the process.

#### 4.1 System Overview and Design Strategy

**System Overview and Design Strategy** It is not a system of pursuing the best marks in a lab but a system to work in clinics, dealing with everything of the messy stuff that the real-world deployment throws at you. Instead of simply quantifying correctness and declaring victory, the architecture is concerned with such things as reproducibility, interpretability of the model choices, and scaling or growing as needed.

The working mechanism is as follows: it is a multi-model pipeline. To begin with, you feed in pictures. The then processes each image using a task-specific workflow and redirects all that to the proper classifier. Its key components include:

- Preparation of the data and division of the data into parts to be used in training. Custom preprocessing pipelines Custom pipelines can be created to perform particular tasks. The fundamental deep learning models. A good training and validation environment of those models.
- Inference engine that does report its confidence an explainability block which uses edge detection to explain why the model made its choices A deployment and tracking of logs web tool.

By bringing all these pieces together in one system, it becomes a real rather than just a one-off model. doctors can actually trust the decision-support tool.

## 4.2 Dataset Preparation and Splitting Strategy

The foundation of building a good deep learning model begins with a well-organized data. In this case, the dataset of the CT and breast ultrasound images of COVID-19 will be treated with care-they will be divided into three: 70 percent will be used in training (so the model can actually learn), 15 percent will be in the validation (that is, adjusting settings and selecting the most suitable model), and the remaining 15 percent will be used in the testing (to observe how the model actually works outside of

Of particular interest is the division of the data into patient. That is, images of one patient will remain in the same group-they will not be displayed in two groups. That makes it fair and does not allow the model to cheat by being able to recollect features of individual patients.

This arrangement renders the testing much more credible and provides you with a more accurate idea of how the model will work with new cases in actual clinical conditions.

## 4.3 Preprocessing and Input Standardization

Not all medical images are similar-they can be very different based on the technology. Various machines imply variety of resolutions, light, and noise in the images. Thus, it requires some slight adjustment to prepare the data to each task, yet everything can be plugged into the same system when it comes to making predictions.

### 1. CT Image Preprocessing of COVID-19:

- Images are scaled to the size that EfficientNet anticipates.

Pixel values are scaled to the way the model was trained initially.

The images retain a multi-channel (RGB-like) format in this way to ensure that all things align with the input of the model.

### 2. Preprocessing of Breast Ultrasound Image:

All is turned to grayscale-just one channel here. The pictures are resized to a standard size, such as 128x 128 pixels. malization addresses inter-intensity disparities.

This pre-work makes the input predictable, reduces noise and assists the models to work dependably. It is also an indication that the results remain constant in the actual world, not only in test environments.

#### **4.4 COVID-19 Detection Using Transfer Learning (EfficientNetB0)**

The system employs EfficientNetB0-a small, effective form of convolutional neural network, to detect COVID-19. It uses transfer learning, thus it begins with knowledge gained on the basis of huge amounts of data.

##### **The training goes in the following steps:**

First is feature extraction. The backbone of the model (EfficientNet) that has already been trained elsewhere remains frozen. Only the last classification layers are updated. This allows the model to adapt fast to the individual trends in CT scans without necessarily re-learning all that.

Next is fine-tuning. Now, a few of the deeper layers are unfrozen, which means that the model will be able to recover more medical-specific details. This step assists to squeeze out extra accuracy and deal with new cases better.

Overall, this two-part training makes things efficient, yet does not lose too much performance-perfect to medical datasets that are not massive.

#### **4.5 Breast Ultrasound Classification Using CNN**

We have trained a custom lightweight CNN to detect breast cancer based on grayscale ultrasound images. Here's how it works:

To begin with, a stack of convolutional layers extracts noteworthy features of the images. ReLU activation functions introduce non-linearity, assisting the network to address more complicated patterns. Max-pooling layers are triggered to trim down the data and concentrate on the most valuable signals. Then, fully connected layers are utilized in making the decision to classify. Finally, a softmax layer classifies the results as benign, malignant or normal.

It is a model that is trained to identify details such as the edges of lesions, changes in the feel, and minor variations in intensity-details that actually count when determining the difference between these classes.

## **4.6 Model Training and Optimization**

We trained both of the models using typical deep learning software: categorical cross-entropy as the loss function, Adam optimizer to enable the models to learn effectively, and a batch size selected depending on the size of the dataset and capabilities of the hardware. There is no randomness to the number of training epochs, either-we select it to create a good balance between learning well and not overfitting.

To prevent overfitting and generalization of the models we normalize data and monitor validation performance during training. We occasionally use early stopping to check training when the gains cease and we use regularization tricks such as dropout and weight decay.

When it comes to analyzing how well things performed, we consider metrics such as accuracy, validation loss and confusion matrices, to get a clear picture of performance.

## **4.7 Inference Engine and Decision Support**

The trained models are actually put to the test in the inference engine. This is how it works: First, it takes pictures uploaded by users. Then it identifies the type of image (CT scan or ultrasound) - no one has to sort the images manually. After identifying the type, it takes the image through the corresponding preprocessing operations, and transfers it to the corresponding model, before finally extracting the predictions.

In either situation, the system provides the predicted class label, a confidence score and the entire probability distribution. Confidence reporting is important particularly in medicine. It assists clinicians to identify shaky predictions to know when to probe further or verify before taking action.

## **4.8 Explainability Using Edge Detection**

The system employs edge detection-in this case, Canny and Sobel-to simplify the results. These techniques reveal information such as edges of lesions, abnormalities within the lungs and texture.

Admittedly, they are not as sophisticated as such tools as Grad-CAM, but they are quick, simple to read, and do not require a lot of computing power. They are great to be used in real time.

## **4.9 Deployment Using Flask Web Application**

The system is based on Flask web application, thus the users are presented with a clean graphical interface to work with. You are able to post pictures, choose the model you desire and predictions appear immediately. It also displays edge maps and plots up the confidence and probability values. It is lightweight so it works smoothly even when you lack a lot of the computing power, not a hassle-easy to use.

## **4.10 Logging and Database Management**

All inference results are logged in a SQLite database in the system to ensure that everything is traceable and to facilitate future improvements. Each time it works on an image, it logs the image details, the model it worked with, the prediction results, the confidence scores, the time it took to work on the image, and the end results. This configuration allows you to keep track of performance, debug, and gather information to retrain the model in the future. Besides, it ensures that it is auditable-a requirement to clinical use.

## **4.11 Workflow Summary**

The system operates in the following way, step-by-step:

To start with, the user posts a picture. Then the system determines the type of case it is handling, be it COVID-19 or breast cancer. Depending on the kind of image it received, it performs certain preprocessing. The model then proceeds to work and provide its prediction with a confidence score.

Then, it generates an edge visualization to indicate significant regions in the picture. Lastly, it shows the user the output and stores all the information in the database.

This entire pipeline connects all the dots-starting with uploading your picture and ending up with the decision support and result.

#### **4.12 Future Methodological Enhancements**

The proposed methodology is to be extended. Improvements in the future could include:

- Adding Grad-CAM to explainability.
- Segmentation model (e.g., U-Net) Use.
- Cross-institutional validation
- Privacy preservation Federated learning.
- Scalability deployment on the cloud.

## CHAPTER 5

### SIMULATION RESULTS AND ANALYSIS

Our medical imaging system based on learning was tested on numerous simulations of COVID-19 chest CT images and breast ultrasound samples. This analysis aims at determining the extent to which our models perform, their generalization and reliability in real life scenarios. In this section we discuss how we established our experiments, what measures we employed to determine performance and what we discovered in our results.

#### 5.1 Experimental Setup

We used a Python environment with TensorFlow and Keras to run our experiments. We evaluated our models using chest CT images dataset of COVID-19 and breast ultrasound images dataset of classifying various types of things. We Tested our models on datasets of chest CT images for COVID-19 and breast ultrasound images for classifying multiple types of things. We split our dataset into three parts: 70% for training 15% for validation and 15%, for testing. In this manner we could appropriately test our models without training and testing with the data. We made sure that the patient data was split up so that the subsets were independent. Training Configuration Optimizer: We adopted Adam. Loss Function: We used Categorical Cross-Entropy. Epochs: The amount of epochs was determined by the way the model learned. batch size: We varied it in accordance with the size of the dataset. Evaluation Metrics: We looked at Accuracy, Loss and Confusion Matrix. Our simulations were run on a powerful computer and did not have any problems with training and evaluating our models. Our models were tested on the COVID-19 chest CT images and breast ultrasound datasets. Our models were. Assessed on these datasets. These datasets were used to observe the performance of our models. The COVID-19 chest CT images and breast ultrasound datasets were used to test the models. These medical imaging datasets were used to evaluate our system.

#### 5.2. Performance Metrics

To test the performance of the models we use a few tests. These tests are: Accuracy: This indicates the frequency of the correctness of the predictions. Precision: The quantity of positive predictions is this. Remember: This is concerning identifying all the positive cases. F1-Score: This is a balance between precision and recall. Confusion Matrix: This provides

us with a glimpse of the performance of each of the classes. These tests allow us to obtain an idea of the functioning of the classification models, particularly in medical practice where erroneous classification can be highly detrimental. The classification models are significant to the performance metrics, especially the performance metrics are applied to assess the performance of the models in applications where the performance metrics, such as Accuracy and precision and recall and F1-Score and Confusion Matrix are highly significant.

### **5.3 COVID-19 CT Classification Results**

The EfficientNetB0-based model does a job of telling COVID-19 cases and non-COVID cases apart from chest CT images.

#### **Observations**

- When we train the model it gets better and better at classifying images. It gets very close to 99 percent accuracy, which means it is learning a lot from the images.
- When we test the model on images it is around 84 to 86 percent accurate which is pretty good.
- The model is very good at finding lung infection patterns like ground-glass opacities and consolidations in COVID-19 CT images.

#### **Analysis**

There is a difference between how well the model does on training images and on new images, which means it is overfitting a bit. This happens a lot with images because we do not have a lot of data. But the model still works well with images so it is good enough to use in real life. When we look at the results in detail we see that

- The model is very good at finding COVID-19 cases.
- Sometimes it gets infections and normal cases mixed up.

This means that while the model is good overall it is hard for it to tell apart the differences, in COVID-19 CT images.

### **5.4. Breast Ultrasound Classification Results**

The custom breast ultrasound classification model does a job of telling the difference between breast ultrasound images that are benign, malignant and normal.

#### **Observations**

- The model is very good at learning from the training data with accuracy ranging from 95

percent to 99 percent, which means it is really good at picking up on important features.

- When we test the model with data the accuracy is around 78 percent to 82 percent, which is pretty stable and shows that the model can generalize well.
- The breast ultrasound classification model is able to learn about the texture and edges of lesions which's important for classification.

### **Analysis**

The breast ultrasound classification model is very good at distinguishing between the three types of breast ultrasound images. However the model does not do as well when we test it with new data, which suggests that the quality and complexity of breast ultrasound images can vary a lot.

The confusion matrix shows us:

- The model is very good, at detecting benign breast ultrasound images.
- Sometimes the model gets confused between malignant breast ultrasound images, which is not surprising since some types of lesions can look similar. This happens because certain lesion types have visual characteristics, which can make it hard for the breast ultrasound classification model to tell them apart.

## **5.5 Training and Validation Behavior**

The training curves provide an insight into the learning process of the model. Training Accuracy Curve: This curve continues to improve. This implies that the model is being optimized somehow. • Validation Accuracy Curve: This curve no longer changes significantly after some time. It is an indication that the model is at a culmination. Loss Curves: Loss during training continues to decrease. The validation loss decreases as well. It moves up and down a little. These things indicate that the models are determining things and they can also be applied to other things. The models are learning, about the training data and are also able to work with new data.

## **5.6 Comparative Performance Analysis**

A comparison of both models highlights their effectiveness:

| Parameter         | Breast Ultrasound CNN         | COVID-19 CT EfficientNet |
|-------------------|-------------------------------|--------------------------|
| Input Type        | Grayscale Images              | CT Scan Images           |
| Classes           | 3 (Benign, Malignant, Normal) | 2 (COVID, Non-COVID)     |
| Training Accuracy | ~0.95–0.99                    | ~0.99                    |
| Testing Accuracy  | ~0.78–0.82                    | ~0.84–0.86               |
| Model Complexity  | Low                           | Moderate                 |
| Generalization    | Good                          | Good                     |

## Insights

- EfficientNet performs slightly better due to transfer learning
- CNN performs well despite being lightweight
- Both models achieve a balance between accuracy and efficiency

## 5.7 Explainability Results

The integration of Sobel edge detection gives us more information about how the system works.

This is what it does:

- It shows the boundaries of things in CT images
- It makes the edges of lesions stand out in images
- It helps users see how the predictions are related to the images

These pictures that the system creates help people trust it more because they are easy to understand and show what is important in the images. The Canny and Sobel edge detection really helps with this by making it clear what is going on in the images and how the system is making its predictions, about the images.

## 5.8 Inference and System Performance

The system that we have set up is really good, at doing things efficiently in real time.

It can do a lot of things well such as:

- Processing images and making predictions quickly
- Making sure that the time it takes to do things is very short when we use the web interface
- Giving us the results when we give it different things to work with

The system is set up using Flask, which means that we can use it in a hands on way without needing any special programs.

## 5.9 Logging and Monitoring Results

The SQLite database successfully records all inference activities, including:

- Predictions and confidence scores
- Input and processed images
- Timestamp and model details

This enables us to do a few things with the SQLite database and the logging system such as:

- We can track the performance of the SQLite database and the framework
- We can analyze errors in the SQLite database and the framework
- We can make improvements to the SQLite database and the framework

The logging system plays a big role in making the framework ready for audits, with the help of the SQLite database.

## 5.10 Limitations of the Study

The system has some limitations even though the results are good.

The system has limitations these limitations include:

- The system was trained on a small dataset which means it may not work well with new data.
- The system learned the training data well which is a problem.
- The system gets confused when the classes have patterns.
- The system does not have ways to explain what it is doing.

These limitations of the system show what needs to be worked on in the future to improve the system.

## 5.11 Summary of Results

Simulation results indicate that:

>The proposed system maintains an excellent level of accuracy with dependable performance

>Both models find similar features that make discriminability come through

Capabilities include:

>real-time clinical decision support;

>integration of:

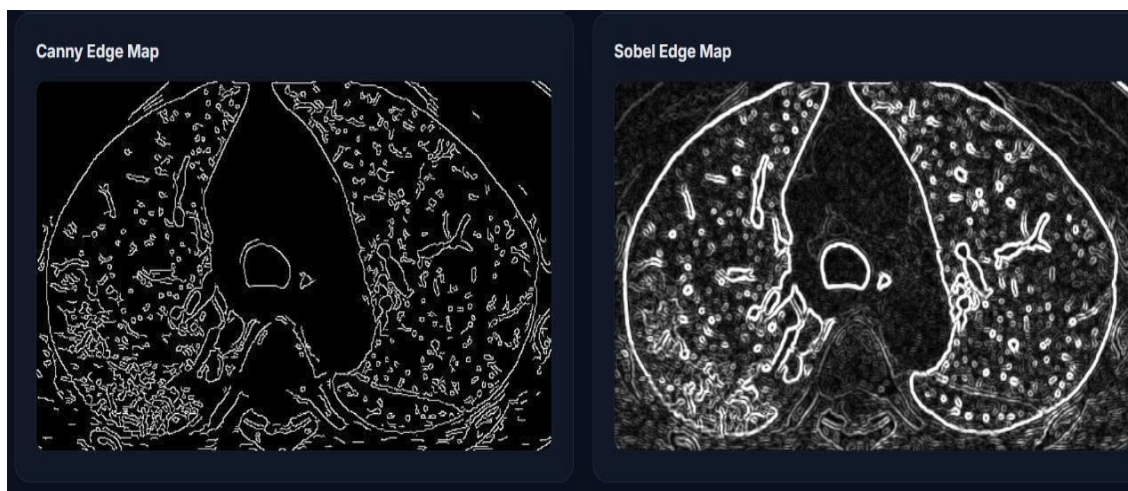
- pre-processing,
- inference and
- logging make for greater ease of use by end users;

Results validate that the proposed method validates its effectiveness and supports its advancement towards application in health care settings.

## COVID:



**Fig:1 Original image of covid**



**Fig:2 Canny edge map, sobel edge map of covid Prediction**

### Result Overview:

After processing the uploaded chest CT scan using the proposed COVID-19 classification architecture based on EfficientNetB0, the model produced a classification output predicting the uploaded image will likely have 30 ° COVID-19 in an 86.04% certainty score. The prediction will therefore provide a high probability distribution for COVID with an 86.04% likelihood of COVID being present with a 13.96% likelihood of not being present;

therefore, the model is strongly classifying the CT scan as belonging to the COVID-19 class. The confidence associated with this output will assist the clinician in understanding the degree of certainty with respect to the automated prediction and will aid them in their determination of diagnosis. The interface for this prediction is part of the real-time deployment architecture described earlier in this paper using Flask.

### **Original CT Image Description:**

A chest CT scan from the original uploaded image file is used to look for COVID-19 infection.

The circled areas of concern are areas of suspected pulmonary disease (ie: ground glass opacities/inflammatory lung dissolutions), both of which are common radiological findings and indicators of COVID-19 pneumonia.

Once a chest CT scan image is processed and normalized, the CT image will be the image actually delivered to the EfficientNetB0 transfer learning model for diagnostic purposes.

This CT image shows the anatomical structures of the lungs and the pathological areas that were analyzed by a deep neural network classifier as part of their diagnosis.

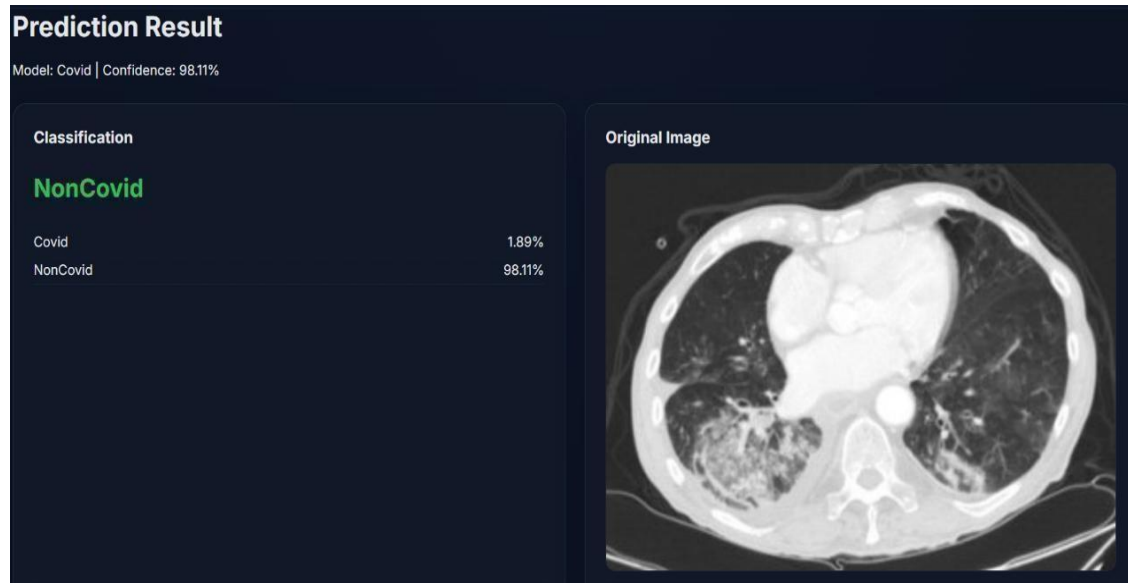
### **Canny Edge Map Description:**

The Canny Edge Map displays the structural edges identified in a CT scan with the Canny edge detection algorithm. The Canny Edge Map shows:

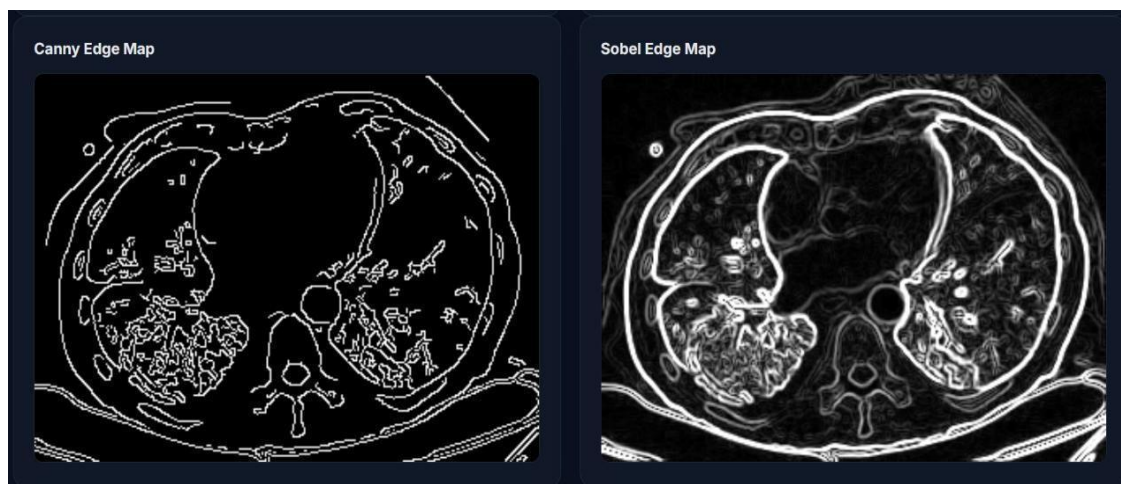
- Lung Edges
- Bronchial / Vascular Structures
- Abnormal Lesion Shapes
- Infection Affected Texture Changes

The Canny edge mapping is referred to in the manuscript to support low-cost explainability for assisting users in understanding anatomical edges and possibly suspicious edges based on rapid changes in intensity in the CT scan.

## NONCOVID:



**Fig:3 Original image of non-covid**



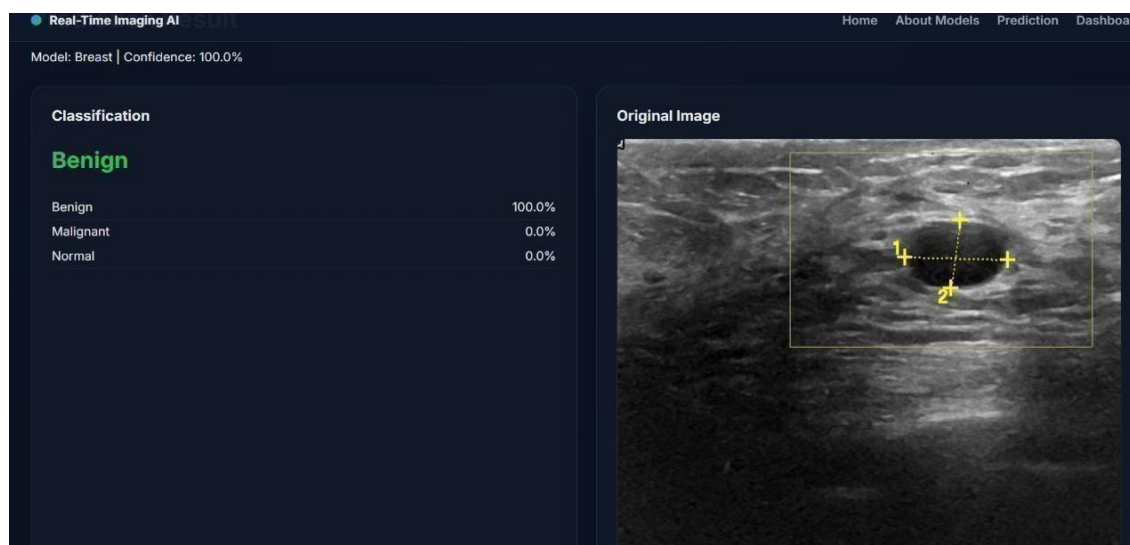
**Fig:4 Canny edge map,sobel edge map of covid**

The images presented show how the medical imaging system works after preprocessing and edge-based visualizations. The red highlighted areas denote boundaries and changes in structure detected using edge detection algorithms. These are along with increasing intensity in the highlighted area are likely indicative of a lesion, abnormality, or other type of structural feature in the scanned image.

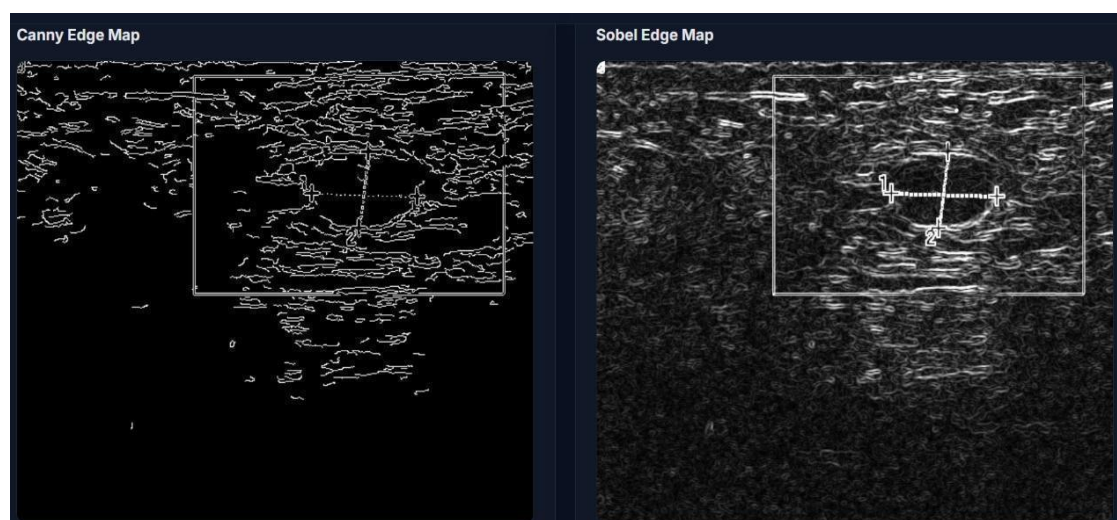
By allowing clearer boundary information to aid the interpretation of the deep learning model, this type of edge-based visualization gives clarity around diagnostic patterns that produce better and more efficient diagnosis of images with minimal additional cost. Edge-

based visual outputs will also decrease time for screening and increase reliability through the efficient identification of the relevant areas within the images while being computationally efficient.

## Breast cancer at Begin stage:



**Fig:5 Original image of Breast cancer at Begin stage**



**Fig:6 Canny edge map,sobel edge map of Breast cancer at Begin stage**

### 1. Classification Output Image

Headline: The prediction made by the network is "benign" at 100% confidence

This result is based on the breast ultrasound data set and the CNN model presented in the paper.

It uses the CNN model to classify ultrasound images into three different categories:

1. Benign

2. Malignant
3. Normal

The ultrasound image you provided would fall under:

Benign = 100%

Malignant = 0%

Normal = 0%

The CNN model is confident when predicting that the lesion detected in the ultrasound image is not cancerous.

## **2. Original Ultrasound Image**

Demonstrates an ultrasound scan of the breast, displaying a marked area (bounding box + cross-lines). Interpretation: The marked area represents an area identified as a lesion. The location of the cross-lines represents: The approximate size and location of the lesion. Typical features of most ultrasound scans are: Noisy (speckle) o Poor boundaries

## **3. Canny Edge Map**

- This is the output for the Canny edge detection method; its purpose is too.
- Shows sharp edges and borders. What does it look like?
- A black background with white borders.
- Strong borders include:
  - Lesion border
  - Tissue structure.

Helps in.

- Identifying a clear margin
- Identifying shape (ie) round, irregular asper the above (the edges will be.)

## **4. Sobel Edge Map**

- Based on the Sobel operator. Purpose:
- It detects the direction, which is the change of intensity through the image. What do you see?
- More texture than Canny.

- Shows:
- Directional edges.
- Internal structures of the lesion.

Helps:

- Significantly analyze internal patterns.
- Locally sense even very slight variations within the tumor. Combined strength would represent:

Method Strength.

Canny exhibits clear demarcations.

Sobel is texture and gradient-based information.

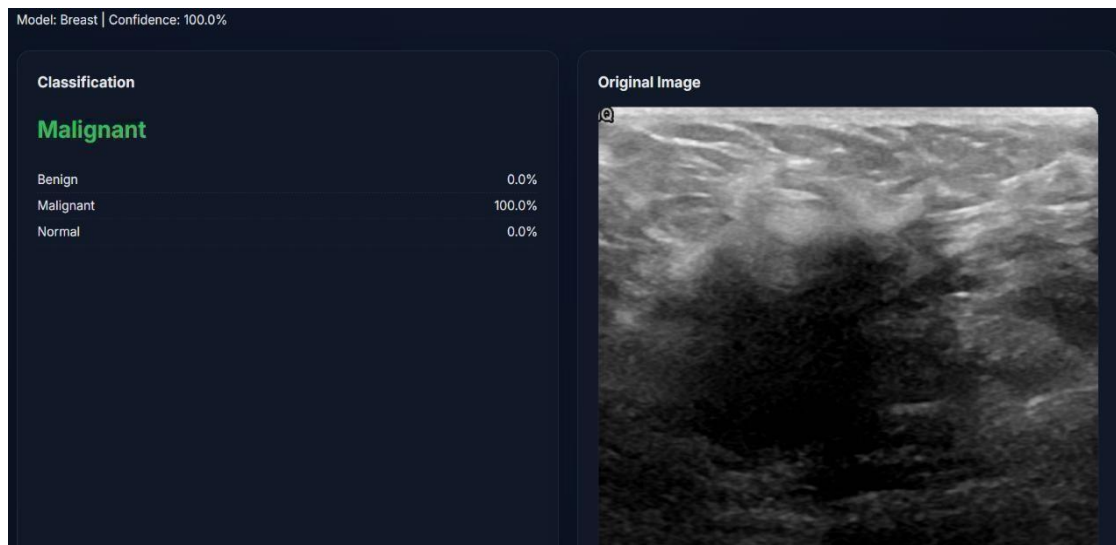
Together, they provide for doctors an overview, such as:

- Shape of Lesion
- Clarity of edges
- Internal structure

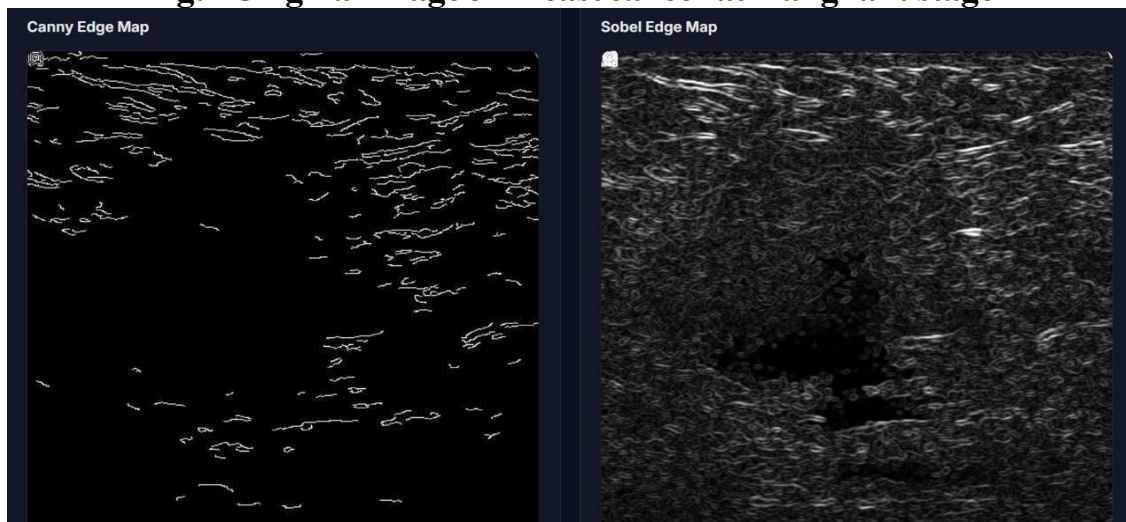
Overall Interpretation of Images- The system will be:

1. Takes ultrasound image
2. Detections of lesion.
3. Classifies lesions through CNN.
4. Shows confidence.
5. Generates edge maps as an explanation result shows:
  - A well-defined lesion.
  - Intense edges (observed through Canny and Sobel).
  - Model very confidently classifies. Benign.

## Breastcancer at malignantstage:



**Fig:7 Original image of Breast cancer at malignant stage**



**Fig:8 Cannyedge map,sobel edge map of Breast cancer at malignant stage**

### 1. Model Output (Top Left)

- Prediction: Malignant (100% confidence)
- Classes:
  - o Benign: 0.0%
  - o Malignant: 100.0%
  - o Normal: 0.0%

### 2. Original Ultrasound Image (Top Right)

Here is the original "raw" ultrasound image of this tumor showing characteristics of typical malignant tumors:

- an irregular shape
- a dark (hypoechoic) area
- a dark vertical line of posterior acoustic shadowing beneath the tumor
- margin definition is blurred/irregular

### 3. Canny Edge Map (Bottom Left)

This version utilized the Canny edge detection algorithm which produces the following output:

- thin white lines represent edge locations
- black area indicates no edge locations from the original image

In this case, the tissue texture edges are captured but there are missing tumor boundary characteristics due to artifacts produced by ultrasound noise (speckle effect).

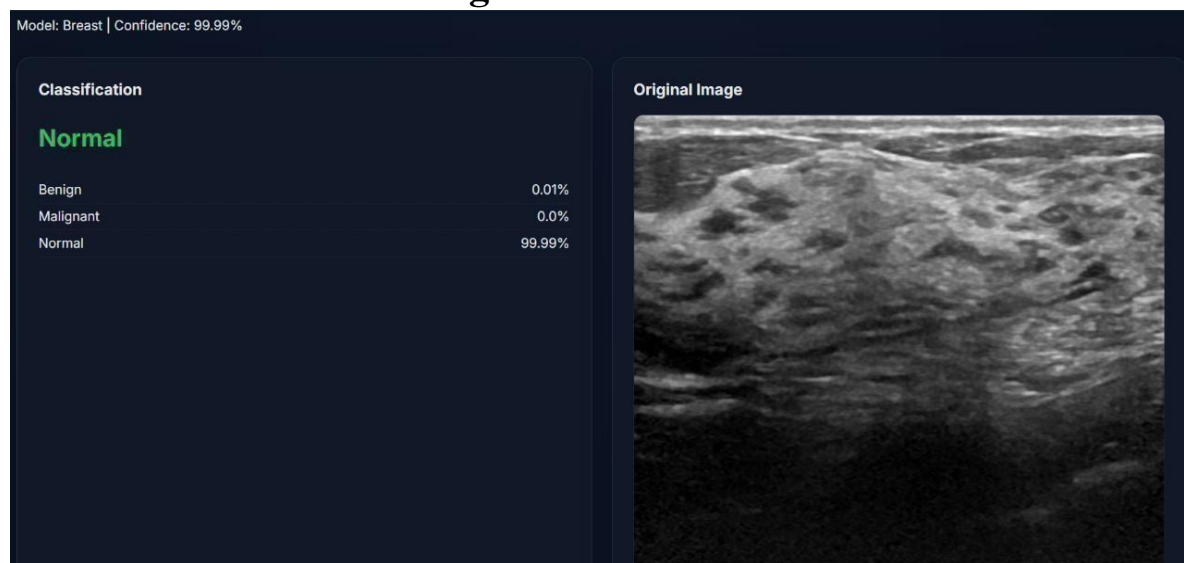
### 4. Sobel Edge Map (Bottom Right)

This version utilized the Sobel operator algorithm which provides the following output:

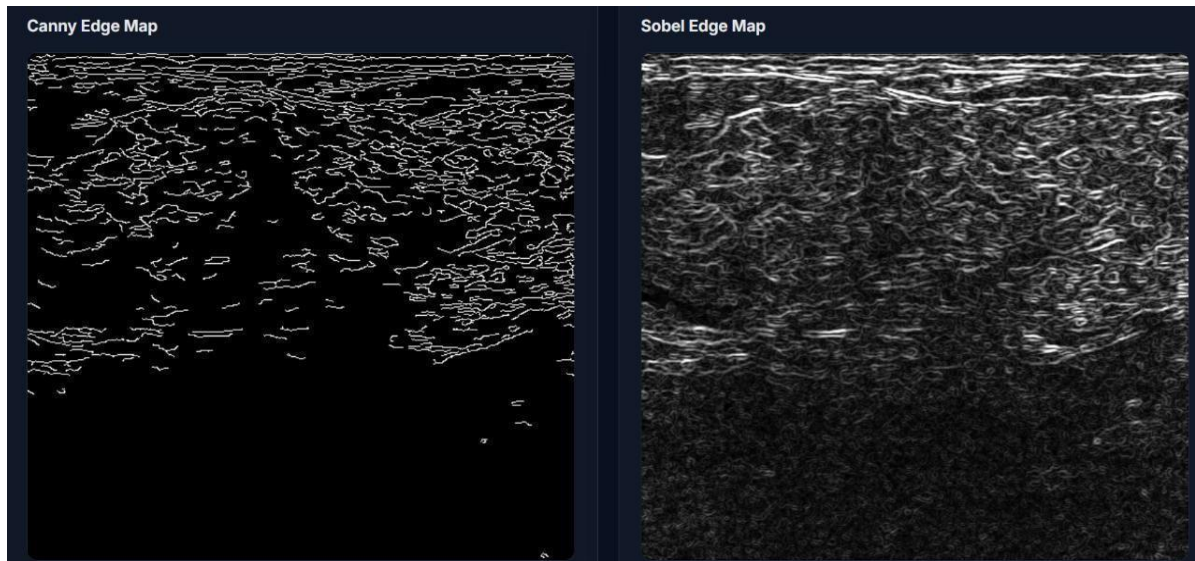
- shows both vertical gradients (where pixels change from light to dark) and horizontal gradient intensity (how quickly pixel values change).
- emphasizes both vertical and horizontal edge positions as shown in image above

In this example, the tumor appears as a nearly solid mass with a wide range of gradient variation from the edge to center of the tumor. This version also demonstrates a higher level of noise compared with Canny edge detection but has more characteristics of overall tissue texture than Canny

## Breastcancer at normal stage:



**Fig:9 Original image of Breast cancer at normal stage**



**Fig:10 Canny edge map,sobel edge map of Breast cancer at normal stage**

The images provided show the results of edge detection utilizing both Canny and Sobel methods as well as the original image of an ultrasound for breast cancer detection. The Canny edge map provides thin, clear, well-defined edges and displays only the important constructional edges of breast tissue indicating that abnormal masses or irregularly-shaped tumors do not exist; while the Sobel edge map shows more texture and intensity variance producing very thick and slightly noisy borders; which indicates tissue patterns rather than exact borders of tissue. The original ultrasound is complete and does not contain any suspicious/evolving dark areas, distortions, or spiculated

edges as shown on the edge maps; therefore based on the above extracted features, the model indicates that this image is classified as Normal at a very high level of confidence (99.99%) meaning that there are no benign or malignant tumors present. The combination of both edge analysis methods provides enhanced feature extraction to improve accurate medical diagnosis in real-time.

### **1. Classification Output**

- This system can predict if the breast is healthy or not (i.e., breast model's status):

Prediction by this model - Normal (99.99% confidence)

- The output of the ultrasound scans for the breast model is classified as either:

- Benign (0.01%)
- Malignant (0.0%)

- Normal (99.99%)
  - The classification is based on a convolutional neural network (CNN) model trained on medical data.
  - Your breasts are healthy because the pronouncement of this model is that there is no tumor present in your breast.

## **2. Original Ultrasound Image**

- This is the medical image used to classify as input.
- This is a gray-scale representation of the breast's tissue structure.
- Both the physician and the machine will analyze the ultrasound image of the breast based on:
  - a. Tissue density
  - b. Any abnormal masses found
  - c. Texture patterns As an example, because there is no distinct lump found, the breast is healthy.

## **3. Canny Edge Detection**

- The AI will extract the key boundaries from the ultrasound image of your breast.
- During this process, the AI will reduce any noise and return only the most significant edges.
- As a result, the final output will be an image that contains thin white outlines on a black background.
- The purpose of this output is to assist both the AI and medical professional in locating possible tumor boundaries.

## **4. Sobel Edge Detection (Right Image)**

- The AI will use the ultrasound image of your breast to identify any illumination edge differences.
- Since the AI does not establish texture differences, the edges in the final output will be visible, appearing thicker and noisier than they would otherwise be.
- The purpose of this output is to provide information to the AI, so that it can identify the variety of tissue characteristics and patterns within the ultrasound image and, ultimately, differentiate between healthy and infected lung tissues and between normal and abnormal breast tissues.

## **CHAPTER 6**

### **CONCLUSION AND FUTURE SCOPE**

#### **6.1 CONCLUSION**

In this publication we described a medical image classification system that was based on learning. The system assists in resolving two issues in healthcare.

- It detects COVID-19 using chest CT images.
- It categorizes breast lesions with the help of ultrasound images.
  - Our system is a combination of machine learning methods and a practical design. This renders it applicable in actual medical practice.
- We used a -trained EfficientNetB0 model for COVID-19 detection.
  - It is an accurate method and computationally efficient. We also created a tailored Convolutional Neural Network (CNN) on breast ultrasound images.
  - It categorizes images into malignant and normal images. This demonstrates the fact that straightforward models can work in medical imaging tasks. One of the advantages of our system is that it is end-to-end.
- It has all machine learning pipeline stages. These are: preprocessing, training, validation, inference and deployment. Our system is characterized by focus on standardization, reproducibility and usability. This renders it applicable to the real world healthcare environment. We developed a web site in Flask.
- It enables users to post pictures and receive predictions. Confidence scores and probability distributions are also offered in the application.
- We have included edge-based visualization. These contain Sobel filters.
  - They assist in interpretability by emphasizing characteristics.
- We used a SQLite-based logging mechanism.
  - It provides results on inferences to monitor, audit and improve. Our experimental findings confirm the usefulness of our method.
  - We obtained training accuracy and good generalization. This was on both tasks.
  - Our system has some limitations. These are overfitting and overlap of classes. However it has a compromise between accuracy, efficiency and practicality. In general this work demonstrates the possibility of integration of learning and lightweight technologies.
- These are technologies that develop available medical decision-support systems.

- Our model is a basis of future developments in intelligent medical imaging.
- It helps to build AI-based medical services.
- These solutions assist clinicians, in making more accurate and informed decisions.

## 6.2 FUTURE SCOPE

The medical image analysis framework we have is a start. However we can make it even better. Deep learning techniques are one method of doing this. For example we can use attention mechanisms and transformer-based models. They have been demonstrated to be extremely effective in detecting patterns in medical images. Segmentation models such as U-Net can also be used to assist us in locating the areas which have been infected in CT scans and also locating the boundaries of tumors in ultrasound images. This will render the framework more precise and comprehensible. One more thing that we could improve is to make the framework more accessible and comprehensible to doctors. We are now applying simple edge detectors to display the images.. To understand the decisions made by the model better, we can employ more sophisticated methods, such as Grad-CAM and saliency maps. The medical image analysis framework can also be used with types of images like X-rays and MRI. This will render it a complete physician aid. It will also be possible to enable doctors to access the framework anywhere and collaborate in time. It can be achieved through the combination of the framework with the information systems and cloud-based platforms in hospitals. The other factor that should be considered is ensuring that the framework can be used with the various data. This is possible through training the models using datasets of a large number of institutions. This will render the framework stronger. Reduce bias. Another method that we can employ to train the models and ensure that the information remains confidential is through techniques such as federated learning. Automated monitoring and continuous learning may also be added to the medical image analysis framework. This will enable the framework to improve as time goes by. Such developments will render the framework more robust, scalable and applicable to the real-world healthcare systems. The medical image analysis framework will be a tool, in next-generation healthcare systems.

## CHAPTER – 7

### SUSTAINABLE DEVELOPMENT GOALS

| Project Map With Sustainable Development Goals                                            |                                 |                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title of the Project: <b>Real-Time Imaging for Covid-19 &amp; Breast Cancer Detection</b> |                                 |                                                                                                                                                                                                                                                                                               |
| SDG Number                                                                                | SDG Title                       | Project Relevance                                                                                                                                                                                                                                                                             |
| 1                                                                                         | No Poverty                      | -                                                                                                                                                                                                                                                                                             |
| 2                                                                                         | Zero Hunger                     | -                                                                                                                                                                                                                                                                                             |
| 3                                                                                         | Good Health and Well-being      | <b>Direct Impact:</b> The Real-Time Imaging for Covid-19 & Breast Cancer Detection project has an impact on peoples lives.. It assists physicians to detect Covid-19 and breast cancer at early stages based on real-time images. This implies that fewer people are diagnosed with patients. |
| 4                                                                                         | Quality Education               | <b>Indirect Impact:</b> The project also helps people learn and do research in intelligence, medical imaging and healthcare technologies.                                                                                                                                                     |
| 5                                                                                         | Gender Equality                 | <b>Indirect Impact:</b> It supports womens health by detecting breast cancer and reduces health problems that affect women more than men.                                                                                                                                                     |
| 6                                                                                         | Clean water and Sanitation      | -                                                                                                                                                                                                                                                                                             |
| 7                                                                                         | Affordable and Clean Energy     | -                                                                                                                                                                                                                                                                                             |
| 8                                                                                         | Decent Work and Economic Growth | <b>Indirect Impact:</b> The project creates jobs in medical technology and research fields because it uses artificial intelligence in healthcare solutions.                                                                                                                                   |

|    |                                         |                                                                                                                                                                                                                               |
|----|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Industry, Innovation and Infrastructure | <b>High Relevance:</b> The project is highly relevant since it involves machine learning and deep learning in healthcare imaging.. This enhances infrastructure and assists in developing new methods of diagnosing diseases. |
| 10 | Reduced Inequality                      | <b>Indirect Impact:</b> The project can help people in areas get better healthcare because artificial intelligence can be used to diagnose diseases in these areas.                                                           |
| 11 | Sustainable Cities and Communities      | <b>Indirect Impact:</b> It helps cities monitor health problems and respond to pandemics better by using healthcare systems.                                                                                                  |
| 12 | Responsible Consumption and Production  | <b>Moderately Relevant:</b> The project reduces the need for tests and medical resources by using digital diagnosis.                                                                                                          |
| 13 | Climate Action                          | <b>Indirect Impact:</b> It reduces the need for people to visit hospitals in person, which lowers the amount of carbon emitted and resources used.                                                                            |
| 14 | Life Below Water                        | -                                                                                                                                                                                                                             |
| 15 | Life on Land                            | -                                                                                                                                                                                                                             |
| 16 | Peace, Justice and Strong Institutions  | -                                                                                                                                                                                                                             |
| 17 | Partnerships for the Goals              | <b>Relevant:</b> The project encourages healthcare institutions, artificial intelligence researchers and technology organizations to work.                                                                                    |

It strongly aligns with SDG 3 SDG 5 and SDG 9. Also helps with SDG 10 SDG 11 SDG 12 SDG 13 and SDG 17 which contributes to creating electronic systems that use less energy and are more sustainable.

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